

FARO AP, Portugal

WMO: 085540

Lat: 37.0167N

Lon: 7.9719W

Elev: 26

StdP: 14.68

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	41.7	44.1	24.0	18.2	52.1	28.9	22.9	53.2	28.1	58.2	25.0	58.4	5.1	20	0.413

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.0	89.5	67.8	86.7	67.8	84.5	67.5	72.6	81.5	71.6	80.5	70.6	79.2	11.0	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.0	110.3	75.5	68.9	106.1	74.9	67.9	102.5	74.4	36.2	81.5	35.3	80.3	34.4	79.2	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.4	20.7	18.4	DB	37.4	96.9	2.6	4.2	35.5	99.9	34.0	102.4	32.5	104.8	30.7	107.8	(4)
			WB	33.7	74.8	3.3	1.3	31.3	75.7	29.4	76.5	27.6	77.3	25.2	78.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	64.8	54.4	55.3	58.7	61.7	66.5	72.2	75.7	76.0	72.0	67.7	60.2	56.2
	DBStd	8.70	3.80	3.73	3.61	4.11	4.97	4.68	4.34	4.26	3.89	3.94	4.34	3.83
	HDD50	16	8	5	0	0	0	0	0	0	0	0	0	3
	HDD65	1399	329	273	196	116	40	2	0	0	1	18	152	272
	CDD50	5410	144	152	271	351	512	665	797	807	660	548	307	196
	CDD65	1319	0	0	2	17	87	217	332	342	212	101	9	0
	CDH74	8275	0	0	8	61	393	1305	2546	2690	1019	247	6	0
	CDH80	2074	0	0	1	4	71	296	737	764	177	24	0	0
Wind	WSAvg	8.2	7.7	8.4	9.1	9.2	8.9	8.8	8.1	7.6	7.5	7.6	7.6	8.3
Precipitation	PrecAvg	19.8	3.1	2.2	2.0	1.4	0.9	0.4	0.0	0.1	0.8	2.4	3.1	3.0
	PrecMax	34.7	12.2	7.6	6.9	4.9	2.7	2.8	0.4	0.9	4.6	7.3	9.0	11.1
	PrecMin	10.5	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.3	
	PrecStd	7.0	2.9	1.9	1.6	1.4	0.7	0.6	0.1	0.2	1.2	2.0	2.4	3.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.2	68.3	74.0	79.8	86.3	90.8	94.8	93.5	87.3	82.8	73.8	68.0
		MCWB	56.5	57.7	58.7	62.2	63.7	66.7	67.1	68.4	67.9	65.6	62.9	60.2
	2%	DB	64.3	65.6	70.2	75.3	81.4	86.2	89.5	89.3	84.3	79.1	71.4	65.8
		MCWB	57.8	57.4	58.1	59.4	63.2	66.3	67.9	68.5	68.0	65.7	62.8	60.6
	5%	DB	62.8	63.7	68.1	72.0	78.3	83.1	86.3	86.5	82.0	76.6	69.5	64.4
		MCWB	57.3	56.8	57.8	59.3	62.4	65.9	67.6	68.6	67.6	65.3	62.4	60.0
	10%	DB	61.5	62.3	66.1	69.5	75.3	80.7	83.9	84.2	79.5	74.4	67.8	63.0
		MCWB	56.3	56.2	57.4	59.1	62.1	65.6	67.4	68.4	67.1	65.4	61.8	58.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.7	61.5	63.3	65.3	69.1	71.7	73.3	74.4	72.9	71.1	68.5	64.1
		MCDB	63.9	64.1	67.8	73.3	77.9	81.1	83.0	82.9	80.3	75.1	71.0	65.3
	2%	WB	60.8	60.1	61.6	63.5	66.9	70.1	71.9	72.9	71.5	69.7	66.3	63.0
		MCDB	62.5	62.9	65.5	70.0	74.6	79.2	82.0	82.0	78.7	73.5	69.0	64.5
	5%	WB	59.2	59.0	60.4	62.2	65.5	69.0	70.8	71.9	70.5	68.7	64.5	61.6
		MCDB	61.3	62.0	64.5	67.8	73.1	78.0	81.0	81.1	77.4	72.6	67.7	63.5
	10%	WB	57.6	57.7	59.4	61.0	64.3	67.9	69.7	70.8	69.5	67.6	63.1	59.9
		MCDB	60.6	61.2	63.7	66.5	71.8	76.8	79.8	80.1	76.2	71.8	66.4	62.3
Mean Daily Temperature Range	5% DB	MDBR	13.4	13.5	13.5	13.7	13.8	14.6	15.8	15.0	13.1	12.3	13.1	12.4
		MCDBR	13.7	14.2	16.6	17.2	18.0	18.4	19.7	18.2	15.9	14.4	13.3	12.1
		MCWBR	9.5	9.5	10.0	9.5	9.2	9.3	10.0	9.5	8.8	8.6	8.5	8.6
	5% WB	MCDBR	10.6	11.9	13.3	13.6	14.0	15.0	16.1	14.9	12.3	10.5	10.5	9.7
	MCWBR	8.9	8.8	9.1	9.1	8.5	8.7	9.6	9.4	8.3	7.7	7.7	8.3	
Clear-Sky Solar Irradiance	taub	0.335	0.352	0.390	0.384	0.393	0.397	0.379	0.399	0.394	0.378	0.350	0.339	
	taud	2.501	2.456	2.347	2.389	2.389	2.395	2.430	2.388	2.424	2.472	2.505	2.497	
	Ebn at Noon	271	279	277	285	282	280	285	277	272	266	263	260	
	Edh at Noon	26	30	37	37	38	38	36	37	34	30	26	24	
All-Sky Solar Radiation	RadAvg	826	1130	1513	1895	2236	2452	2511	2234	1763	1253	902	734	
	RadStd	89	122	134	121	141	118	62	67	97	95	81	75	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+137
	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page